

SAYAN GOSWAMI

MSc in Data Science and AI, RKMVERI, Belur

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GitHub
LinkedIn
Portfolio



SUMMARY

My experience combines a foundation in Machine learning, Deep learning, Multimodal AI & RAG's with hands-on expertise in building end-to-end intelligent systems. I have worked on projects spanning healthcare, Generative AI and LLM evaluation, covering data preprocessing, model development, deployment, and performance assessment. My current research focuses on Dynamic LLM Prompting with RAG and Cross-Attention for patient-specific chest X-ray disease classification, while my broader interests include Gen AI, RAG's and real-world AI applications.

EDUCATION

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)
MSc in Data Science and Artificial Intelligence, 2025–27

SGPA (2nd Semester) – 6.00

Bankura Christian College, Bankura
BSc (Honours) in Mathematics, 2022–25
CGPA – 7.72

DAV Model School, Durgapur
Class XII (CBSE), 2022
Percentage – 87%

EXPERIENCE

Research Intern - Indian Institute of Technology, Patna

Supervisor: Dr. Joydeep Chandra
May 2026 – July 2026

Worked on Dynamic LRC-CXR, a retrieval-augmented prompting framework for patient-specific chest X-ray disease classification, integrating Mistral-7B, CheXzero, and BioClinical-BERT for dynamic medical knowledge retrieval and multimodal image-text fusion.

SKILLS

- **Languages:** Python, C, SQL
- **Frameworks:** NumPy, Pandas, Matplotlib, Scikit-Learn, PyTorch, LangChain
- **Operating Systems:** macOS, Ubuntu Linux
- **Tools:** Git/GitHub, VS Code, Excel, Google Colab

COURSE WORK

- Machine Learning, Artificial Intelligence, Deep Learning, LLM's and Agentic AI
- Probability and Statistics, Linear Algebra
- Distributed Computing and Data Structures and Algorithms
- Database Management System

Achievements & Certificates

- NPTEL – The Joy of Computing using Python (2026)
- CUET-PG 2025 (Mathematics) – Qualified

PROJECTS

NādaAI - Deep Learning Based Music Generation

RKMVERI

Jan 2026 – Apr 2026

Developed an end-to-end deep learning pipeline that converts 5 seconds of vocal humming into structured musical melodies. Integrated CREPE-based pitch detection, MIDI sequence generation, and a two-layer LSTM architecture trained on approximately 495,000 musical note sequences. Implemented real-time audio synthesis and a Streamlit interface, enabling melody generation without requiring MIDI input or formal music knowledge.

Customer Churn Prediction Using Machine Learning

RKMVERI

Oct 2025 – Dec 2025

Developed an end-to-end machine learning pipeline to predict telecom customer churn using demographic, billing, and service usage data from 7,043 customers. Implemented preprocessing, SMOTE-based class balancing, model selection, hyperparameter optimization, and performance evaluation. Deployed the final tuned Random Forest model as an interactive Streamlit dashboard providing churn risk prediction, feature importance analysis, and business decision support.

RAG Evaluation Framework

RKMVERI

Feb 2026 – May 2026

Conducted a literature survey of RAG evaluation methods, covering retrieval and generation-phase metrics. Built a Retrieval-Augmented Generation (RAG) chatbot (Mistral + FAISS + LangChain) with automated quality evaluation using RAGAS metrics (faithfulness, answer relevancy, context precision/recall, answer correctness) judged by LLaMA-3.3-70B. Added a CI/CD quality gate via pytest and GitHub Actions that blocks pushes when scores fall below threshold.

PriceScope - Distributed Price Comparison Platform

RKMVERI

Jan 2026 – May 2026

Designed and developed a distributed price comparison platform using FastAPI, Celery, and Redis, aggregating real-time product data from Amazon, eBay, Google Shopping, Walmart, and Target with fault-tolerant microservices. Integrated MCP with GPT-4o-mini for AI-powered product insights.

ACADEMIC RESEARCH

Mathematical Modeling Using Calculus

B.Sc. Dissertation

Feb 2025 – Jun 2025

Studied mathematical modeling of dynamic systems using calculus, focusing on optimization, logistic population growth, epidemic modeling, and engineering applications.

HOBBIES & INTERESTS

Singing, Poetry, Fitness